

CASE STUDY: Toyota's Disrupted Global Supply Chain, COVID-19 and Global Chip Shortages

The case examines how companies can manage supply risks for black swan events such as the global microchip shortage that Toyota and other automakers faced during the coronavirus pandemic. We have an opportunity to contrast the principles of “lean” approach for core process efficiency and the added needs, and supporting concepts, in building “resilience” for future unexpected shocks.

ASSIGNMENT QUESTIONS:

1. What are some of the unique attributes of the Toyota Production System, and what are the implications for Toyota's supply chain? In what ways these aspects of the Toyota's system and supply chain might make it vulnerable to unexpected (“black swan”) events like the 2011 Earthquake in Japan, and more recently the global COVID pandemic? Explain why and provide examples on how they affected the Toyota responses in the above mentioned crises. **(TEAM 1)**
2. What did Toyota learn from the 2011 earthquake and tsunami? What changes did the company make to create a more resilient supply chains? Stating it in different words, what practices did they put in place after the 2011 disaster to create future resilience? Did they result in substantial deviation from the “lean practices”? **(TEAM 2)**
3. Did the changes that Toyota made after the earthquake in its supply chain help with the global chip shortages in 2020 and 2021? How would you compare Toyota's performance during the pandemic, and especially during the chip shortages, to the other automakers? Would you call the Toyota supply chain resilient in its performance during the 2020-2022 window? **(TEAM 5)**
4. What were interesting practices that other automakers used to deal with global chip shortages? How would you adapt these practices to prepare to deal with future shocks in other causes and materials? What would you advise as the main elements for building “resilience” in global supply chains (if it helps you narrow it down to a specific industry you work or understand through your readings) in anticipation of unknown nature future shocks? What are the important future risks you are interested in building resilience for? **(TEAM 4)**